NAME:KRISHNAPRIYA P

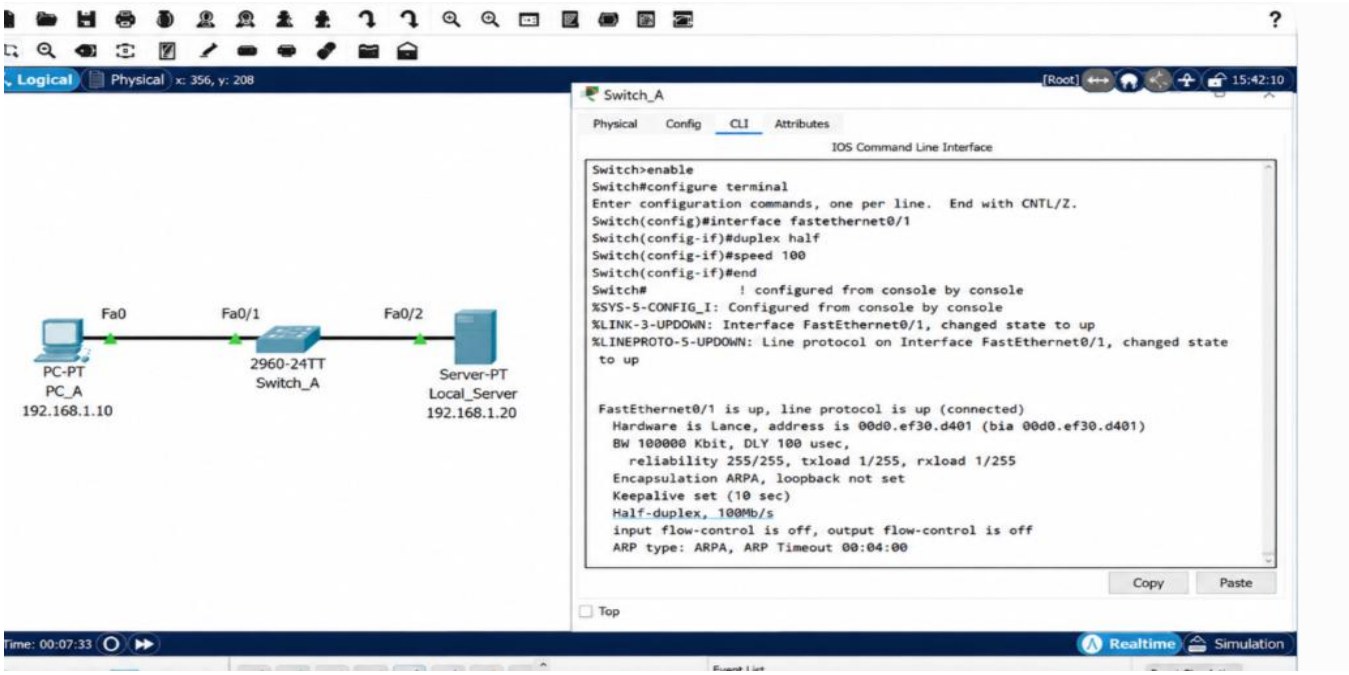
REGNO:192511279

COURSE CODE:CSA0712

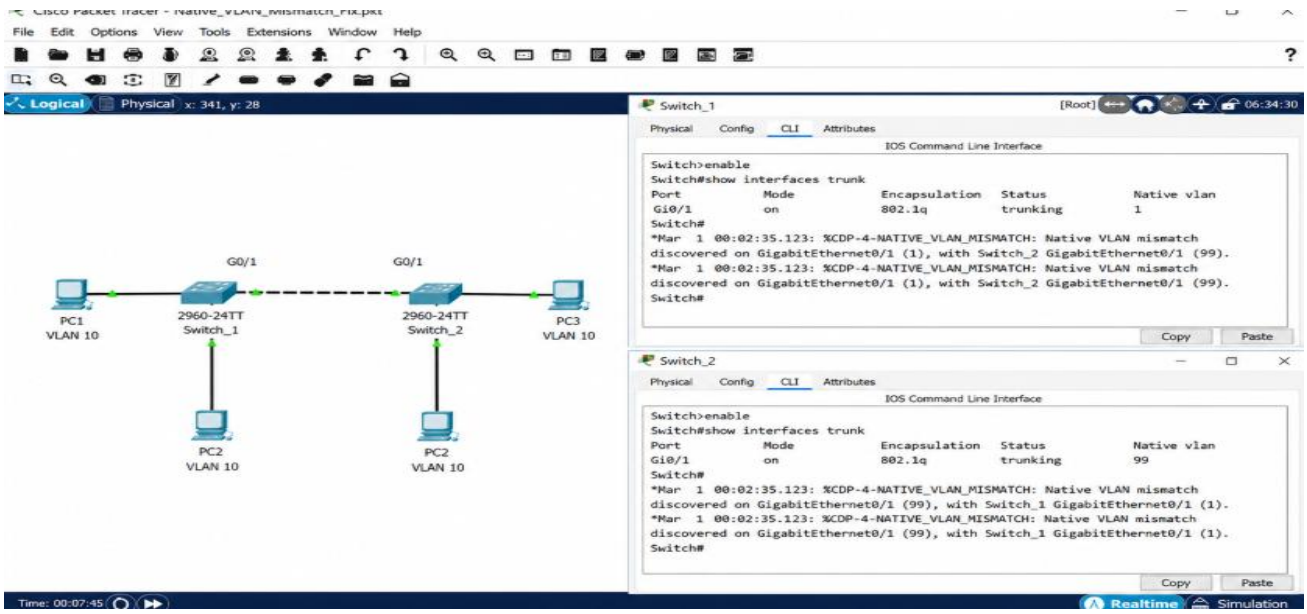
COURSE NAME:COMPUTER NETWORKS

CO2-AT3- Debugging and Optimization

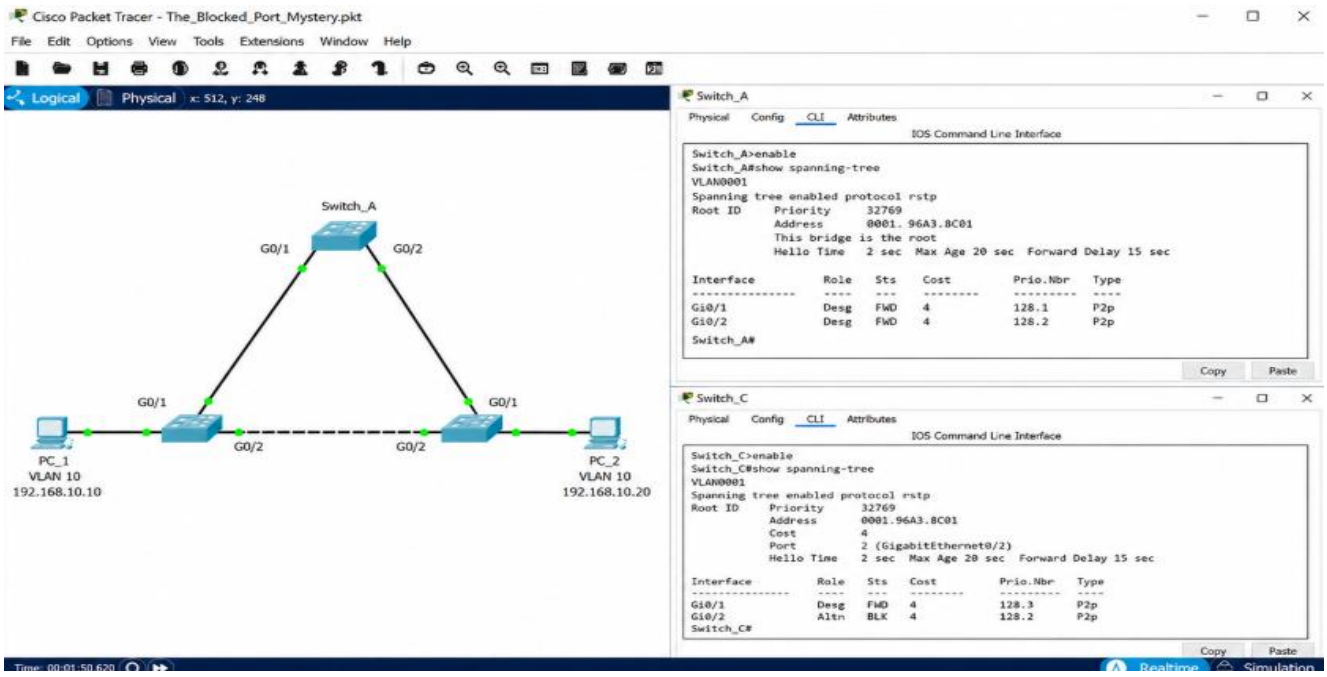
1. The Half-Duplex Bottleneck



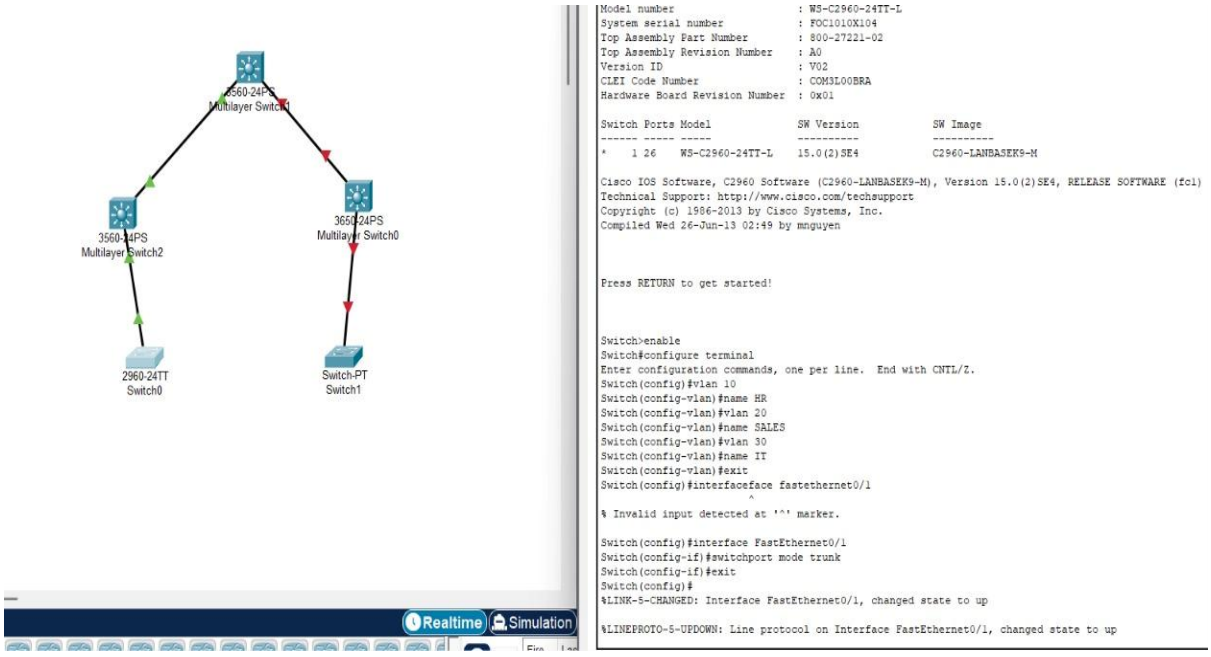
2. The Native VLAN Mismatch



4. The Blocked Port Mystery



6. Root Bridge Optimization



7. EtherChannel Link Aggregation




```
Switch(config)#enable
% Incomplete command.
Switch(config)#configure terminal
                ^
% Invalid input detected at '^' marker.

Switch(config)#interexit
                 ^
% Invalid input detected at '^' marker.

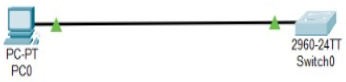
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------


1      Po1(SD)           LACP   Fa0/23(I) Fa0/24(I)
Switch#
```

8. PortFast and BPDU Guard Implementation [1]




```
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if)#spanning-tree bpduguard enable
Switch(config-if)#exit
Switch(config)#show running-config
                ^
% Invalid input detected at '^' marker.

Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show running-config
Building configuration...

Current configuration : 1160 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode access
 spanning-tree portfast
 --More--
```